

# Delivery Allocation using BFS and Genetic Algorithm

## 1. Introduction

This project addresses a **delivery allocation problem** where multiple delivery points must be assigned to two vehicles. The primary objective is to achieve **load balancing** by minimizing the difference between the total travel distances of both vehicles.

To solve this problem, we combine:

**Breadth-First Search (BFS)** for shortest path calculation

**Genetic Algorithm (GA)** for optimization

## 2. Problem Description

The environment is represented as a **10×10 grid** that includes:

A **Depot** (starting point)

Multiple **Delivery Points**

Several **Obstacles**

Each delivery point must be assigned to one of the two vehicles in a way that balances the total distance traveled.

## 3. Environment Representation

The grid is modeled as:

0 → Free cell

1 → Obstacle

The depot is located at position (0,0), and delivery points are distributed across the grid.

This structure simulates a realistic environment with blocked paths.

## 4. Search Algorithm (BFS)

### Algorithm Used

### Breadth-First Search (BFS)

### Why BFS?

Guarantees the **shortest path** in an unweighted grid

Explores nodes level by level

Works effectively with obstacles

| Feature       | BFS                  | DFS         |
|---------------|----------------------|-------------|
| Shortest Path | Yes                  | No          |
| Traversal     | Level-wise           | Depth-wise  |
| Suitability   | Distance calculation | Exploration |

## 5. Genetic Algorithm Design

### 5.1 Chromosome Representation

Each solution is represented as a binary chromosome

[0, 1, 0, 1, 0, 1]

0 → Vehicle 1

1 → Vehicle 2

## 5.2 Fitness Function

$$\text{Fitness} = 1 / (1 + |V1 - V2|)$$

Where?

V1 = total distance of vehicle 1

V2 = total distance of vehicle 2

### Why this function?

Converts the problem into a maximization problem

Best case → fitness = 1 (perfect balance)

Worst case → fitness approaches 0

## 5.3 Selection Method

### Tournament Selection

Why?

- Selects the best individuals from random subsets
- Reduces weak solutions
- Efficient and simple

## 5.4 Crossover Method

### Single-Point Crossover

Why?

- Suitable for binary representation
- Combines parent solutions effectively

## 5.5 Mutation Method

### Bit-Flip Mutation

Why?

- Flips bits ( $0 \leftrightarrow 1$ )
- Introduces diversity
- Prevents convergence to local optima

## 5.6 Termination Condition

The algorithm stops after a fixed number of generations:

Generations = 300

**Why?**

- Controls execution time
- Allows sufficient evolution of solutions

## 6. Results

The algorithm outputs:

- Best chromosome (final assignment)
- Fitness value
- Total distance for each vehicle

The result demonstrates a **balanced distribution** of delivery tasks between the two vehicles.

## 7. Visualization

### 1. Grid Map

Displays:

- Depot
- Delivery points
- Obstacles
- Vehicle assignments

### 2. Fitness Graph

Displays:

- Fitness improvement across generations

#### **Why Visualization is Important?**

- Helps understand the solution
- Verifies correctness
- Shows convergence behavior of the algorithm

## 8. Conclusion

This project demonstrates how combining:

- A search algorithm (**BFS**)
- An optimization technique (**Genetic Algorithm**)

can effectively solve real-world problems such as delivery allocation.

The approach ensures:

- Accurate distance computation
- Efficient load balancing

## 9. Future Improvements

- Use A\* instead of BFS for better performance
- Increase the number of vehicles
- Add real-world constraints such as traffic or time windows

